

Waste Classifier App

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SODA Mentorship Program

Presentation

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Background and Inspiration



ASU's OSCAR Sort

- Problem: Lack of awareness and transparency around where waste goes after you dispose of it, leading many to unintentionally increase excess waste sent to landfills
- Inspired by **Arizona State University's OSCAR** waste sorting systems around campus, I designed the **Waste Classifier App** to help ASU students and faculty properly dispose of their waste
- Advantage: **Easier to use and available all the time and everywhere**, as compared to limited locations of ASU's OSCAR model
- Aligns with ASU's goals of making more **sustainable** and environmentally-friendly choices

Waste Classifier App

Web Application where users can upload a picture of their waste and be informed of how to properly dispose of it, including locations on ASU's campus.

Waste Classifier App

Enter your Image



Drag and drop file here

Limit 200MB per file • JPEG, JPG, PNG

Browse files



biological137.jpg 11.7KB

Classify

The image has been uploaded.



Learn about where to throw your waste!

This image is classified as Organic Waste.

Organic waste such as food scraps and garden trimmings decomposes naturally and can produce nutrient-rich compost. Correct disposal reduces landfill methane emissions and supports soil health when composted. Mixing organics with other trash contaminates recycling streams and contributes to greenhouse gas emissions.

Organic waste should go in compost bins or organic waste collections. This includes fruit and vegetable peels, coffee grounds, and garden waste. Avoid plastics or non-compostable materials in these bins.

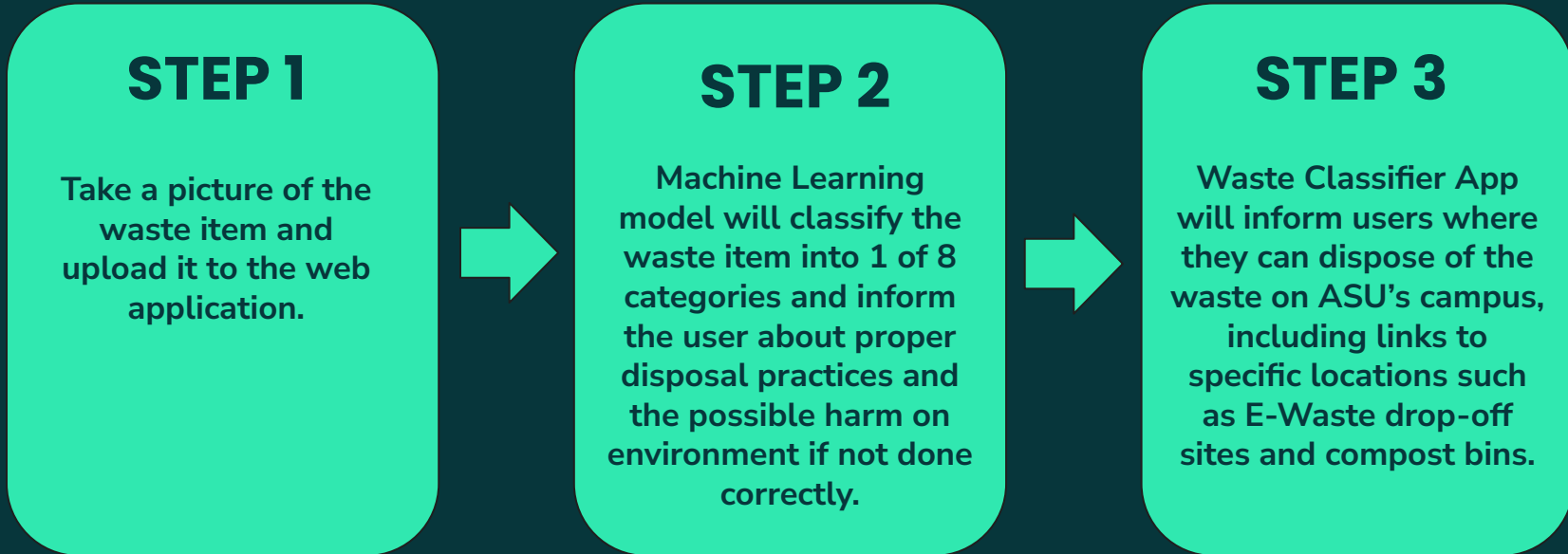
Arizona State University Waste Information

Organic items, such as leftover food, can be placed in any compost container on campus. Locations include:

1. Memorial Union - Food Court and Plaza : <https://www.google.com/maps?&daddr=33.41777,-111.93438>
2. Hassayampa Dinning Hall: <https://www.google.com/maps?&daddr=33.41626,-111.92868>
3. Sun Devil Athletic Stadiums
4. Residential halls
5. Community kitchens and dinning halls

Waste Classifier App

Web Application where users can upload a picture of their waste and be informed of how to properly dispose of it, including locations on ASU's campus.



Waste Classifier App DEMO:



Methodology

Step 1: Machine Learning Classification Model

- Gathered “New Trash Classification” Dataset from Kaggle
- *Split Data into 8 Categories:* Paper, Plastic, Glass, Cardboard, Metal, Organic, E-Waste, Textile, Landfill
- Used Google’s Teachable Machine interface to create a Machine Learning model based on **Convolutional Neural Network** architecture
- **Keras** library used for deep-learning framework in the ML model due to its easy-to-use interface for training neural networks and support of sequential models for stacked layers
- Set Model’s parameter values for optimal accuracy after experimentation
 - Epoch: 50
 - Batch size: 16
 - Learning rate: 0.001
- Trained and tested AI model for highest accuracy and efficiency

Methodology

Step 2: Web Application with User Interface

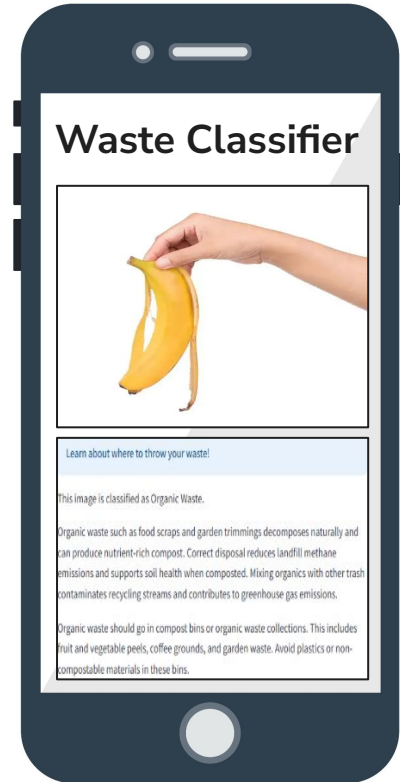
- Create web application using Streamlit
 - **Streamlit** chosen as it allowed for an efficient open-source Python framework to create an interactive data application for the machine learning model
- **Python** programming language
- Libraries
 - Tensorflow: chosen because it can run the AI/ML models on CPU efficiently for creating the web application
 - Numpy: chosen because it allows for efficient implementation of multi-dimensional arrays of images and for mathematical operations in model's classification predictions
 - Pillow: chosen because of ease to work with pictures in many formats, including transforming and resizing the test and train images for the model

Future Plans

In the future, I would like to turn the Waste Classifier App from a web application to a mobile application to increase ease of use for students and faculty. Through the mobile application, users can directly take a photo from the app and get instant information regarding proper disposal methods and locations at ASU, without the extra step of having to upload the photos.

A feature I would add to the Waste Classifier mobile application is that ability for the program to output the closest disposal (recycle, trash, compost, etc) bins available to a person via access to the person's location.

Mobile App
Waste Classifier
Design



Mentorship Program

Mentor: Aaditya Jindal
Mentee: Sargun Bhatia

My experience with mentor Aaditya Jindal:

- Kept me on track to complete the project with weekly meetings where we discussed what had been accomplished and what was left to do.
- Provided guidance on the Artificial Intelligence and Machine Learning components of the project, including providing a step-by-step process of how machine learning models are created and guiding me through each step week by week.
- Explained his past work with machine learning models as examples of the process I should follow, including how to prepare the dataset, create model layers, set components such as batch size and learning rate, and train the model.
- Highlighted different resources and research papers where I could get additional information for my project.



Thank You!